



Day 3: Data Cleaning with Power Query

Masterclass Series: Learn the core transformation workflows to clean and prepare raw business data effectively.

The Analytical Dilemma

⚠ Messy Data Reality

A company provides you with raw sales spreadsheets containing:

- ✗ Blank or null sales records
- ✗ Duplicate transaction IDs
- ✗ Inconsistent spellings and formats

📈 The Core Rule

Can we directly build executive dashboards on this messy raw data?

Absolutely NOT. Messy inputs lead to broken, inaccurate reports.

"First we clean data. Then only we analyze it."

What is Power Query?

The ETL Workhorse

Power Query is the dedicated, visually-driven data transformation and cleaning engine built directly inside Power BI.

Primary Capabilities:

-  Stripping and filtering raw rows
-  Standardizing inconsistent formats
-  Fixing spelling and database errors



Power Query Interface Overview

Power Query Editor - Student_Performance_Dataset				
Home Transform Add Column View				
QUERIES (1) Student_Data	StudentID		Marks	City
	101	85	Hyderabad	
	102	<i>null</i>	hyderabad	
APPLIED STEPS Source Navigation Cleaned_Rows				

The Navigation Framework

Access Power Query via: **Home** → **Transform Data**

-  **Queries Pane (Left):** Quick lists of active source tables.
-  **Data Preview (Center):** Interactive preview grid of active columns.
-  **Applied Steps (Right):** Automatically records every cleaning step.

Removing Duplicates & Nulls

Ensuring Analytical Integrity

Duplicate Rows: Distort dashboard totals and metrics.

Method: Select key column → **Remove Rows** → **Remove Duplicates**

Null Values: Represent empty data cells. Can crash metrics.

Method: Right-click column → **Replace Values** (e.g., replace *null* with 0)

Raw (Duplicates/Nulls)

StudentID	Name	Marks	
103	Raj	92	
103	Raj (Dup)	92	
104	Kiran	null	

Cleaned Out

StudentID	Name	Marks	
103	Raj	92	
104	Kiran	0	

Data Types & Column Naming

Fixing Data Types

Wrong datatypes cause visual and calculation errors.

Supported Formats: Text, Whole Number, Decimal Number, Date.

Step: Select Marks Column → Change type from **Text** → **Whole Number**.

Descriptive Columns

Descriptive naming boosts dashboard readability and metric development.

Step: Rename column **Attendance** → **Attendance_Percentage**.

Renaming columns in Power Query helps keep your data model organized.

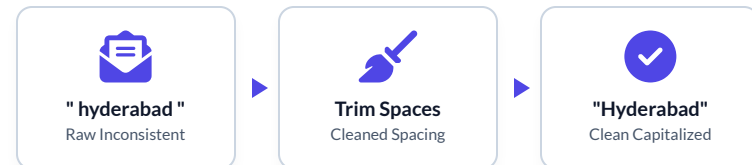
Basic Transformations & Shaping

Standardizing Raw Text

- ✂ **Trim:** Strips leading and trailing white spaces.
- H Capitalize:** Standardizes inconsistent casings (e.g., *hyderabad* → *Hyderabad*).
- ✂ **Split/Merge:** Splits full names or concatenates code fields easily.

Data Shaping

Removing unneeded columns, filtering rows, and organizing columns keeps reports optimized and running quickly.



Interactive Live Demo

☰ Part A: Data Extraction

- 1 Open Power BI Desktop and choose: **Get Data** → **Excel**
- 2 Select:
Student_Performance_Dataset.xlsx
- 3 Click the **Transform Data** button to launch Power Query

✂ Part B: Cleaning & Formatting

- 1 Remove student duplicates using **StudentID**
- 2 Replace null marks with **0**
- 3 Capitalize **City** and change marks column to **Whole Number**
- 4 Rename column and hit **Close & Apply**

Hands-on Practice Tasks

Student Instructions

Open Power Query on your machine and complete these data-cleaning tasks on the raw dataset:

- ✓ Task 1: Find and delete duplicate students
- ✓ Task 2: Spot null sales cells and replace them with 0
- ✓ Task 3: Standardize lowercase city spelling variations
- ✓ Task 4: Match and set correct whole number datatypes

Discussion Questions

Keep these real-world data principles in mind during the exercise:

- ? "Why are duplicate rows dangerous to final totals?"
- ? "Why do datatypes matter before writing dashboard metrics?"
- ? "What happens if corporate database sources are loaded messy?"

20-Min Analytics Challenge

The Clean Dashboard Build

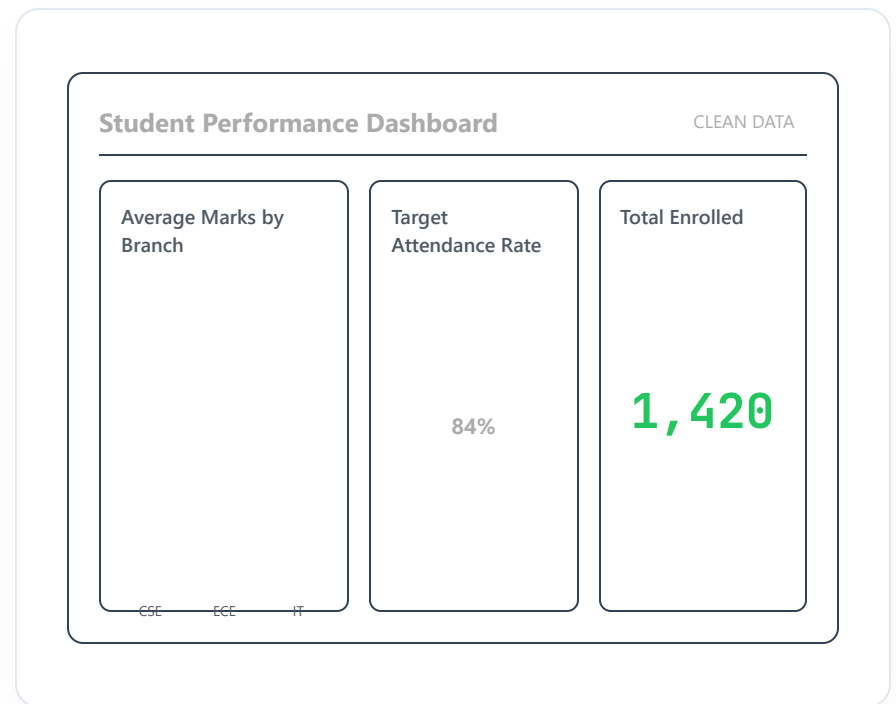
Once your data is cleaned and shaped in Power Query, load it to build a student performance dashboard:

 **Visual 1:** Average Student Marks by Branch Chart

 **Visual 2:** Attendance Rate Analysis Gauge

 **Visual 3:** Student Enrollment Count by City Map

Can you present a clean, functional student performance dashboard before the timer ends?



Exam Guide & Industry Scope

Exam Study Guide

-  **Power Query:** Core tool used for data cleaning and transformation.
-  **Applied Steps:** Records transformations automatically to keep refreshes repeatable.
-  **Duplicates & Nulls:** Must be cleaned to prevent visual rendering bugs and calculation errors.

Sector Applications

-  **Banking:** Standardizing transaction records.
-  **Healthcare:** Resolving duplicate patient profiles.
-  **E-Commerce:** Normalizing complex online product lists.
-  **HR:** Cleaning employee shift records.

Common Doubts & Session Wrap-up

Standard Student Doubts

Q: Why delete duplicates instead of keeping all logs?

 Duplicates skew metrics and average calculations.

Q: What makes Applied Steps so useful?

 It automates your data prep, running all cleaning steps on refresh.

The Data Standard

"Even the best-designed dashboards are useless if the underlying data is wrong."

Key Takeaway: Raw data is messy. Analysts bring order, clarity, and meaning to it.